The de.NBI Cloud Federation in 2025

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Overview

To meet the rising demand for comprehensive bioinformatics tools in the life sciences, the **German Network for Bioinformatics Infrastructure** (de.NBI) was established in 2015 as a national bioinformatics consortium offering high quality bioinformatics services, comprehensive training and powerful computing capacities. The de.NBI Cloud is one of the flagship services for the **de.NBI & ELIXIR Germany** network with more than 4000 registered users, over 1350 projects, of which more than 620 are currently active, and over 1380 publications citing its use.

Compute Power for Your Research

Eight federated cloud centers adhere to a **unified governance structure** and utilize the centralized project application and management workflow available through the **de.NBI Cloud Portal**. Single sign-on and user management are provided via **LifeScience Login** and **eduGAIN**. PIs from German universities or research institutes may submit project applications, which are reviewed by the Cloud Access Committee and, upon approval, resources are allocated to one of the cloud centers. Regular staff training, mutual internal audits and external **ISMS certification** (ISO27001 or BSI C5) ensure secure operations of the data centers.

Project Types for All Skill Levels

The de.NBI Cloud offers various project types for your computing needs, covering different levels of expertise and experience in cloud computing: From the simple and fast start of pre-configured VMs with easy access (**SimpleVM**), to the use of bioinformatics tools and workflows (**Galaxy**), to the possibility of running your own infrastructures and services (**OpenStack & Kubernetes**).

Virtual Research Environments

We offer pre-configured **virtual research environments** (VRE), which can be accessed easily when using our **SimpleVM** platform. They include **integrated development environments**, **data science notebooks** and **virtual desktop environments**. The VREs can be accessed in the web browser or by secure shell (ssh). VREs and images tailored for specific needs can also be used to provide uniform VMs for **trainings and workshops**. The cluster module makes it easy to set up a SLURM-based **grid cluster** for resource-intensive computations. Users can also provide recipes for custom VREs.

Resources for NFDI and EOSC

The **de.NBI Cloud** provides computational resources for a diverse portfolio of projects within Europe (ELIXIR, EOSC, Horizon Europe) and Germany (**NFDI** projects), ranging from (web)service development and **hosting**, to execution of memory and CPU-intensive bioinformatics **workflows**, **GPU** resources for **machine learning**, resources for **hackathons**, **workshops and trainings**, as well as resources for proof-of-concept development, and secure environments for **sensitive data processing**.

Training Matching your Expertise

Regular trainings and workshops teach you how to use the cloud infrastructure effectively for your research. The annual **Cloud User Meeting** offers users the opportunity for **networking** and to get in touch with the administrators of the cloud locations and the cloud governance. Researchers **present their work** in the cloud and **training courses** expand the range of expertise for experienced cloud users, while also providing a comprehensive introduction for beginners.

Selection of training course topics:

- VM basics: Linux & command line
- SimpleVM & Research Environments
- BioConda & BioContainers
- CWL, Nextflow, Snakemake
- Infrastructure as Code: Ansible, Terraform
- Grid computing: BiBiGrid / SLURM
- IaaS / PaaS: OpenStack, Kubernetes
- Using Galaxy Tools and Workflows

CLOUD USER MEETING

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